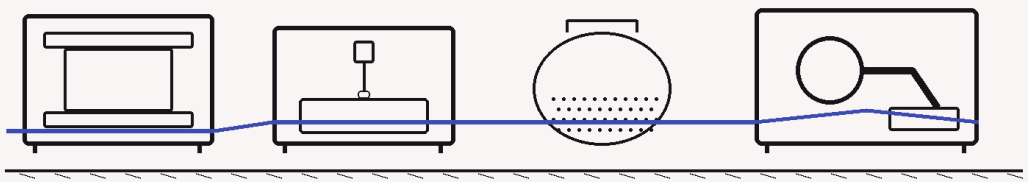
Before a lab spends a year on a material, know whether to trust the prediction.

Batteries, catalysts, frameworks, climate materials, alloys. Machine-learned potentials screen them faster than labs can test them. We measure where these models go wrong, against published references and gates set in advance, and publish every verdict.



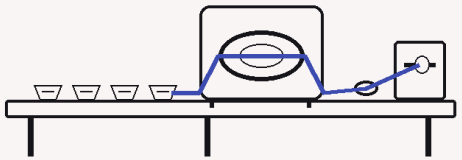
Lupine Science, company facts (lupine.science/about/), data/company.json.

380,000 computationally stable structures; 736 independently synthesized by late 2023: a reported 0.2% validation rate.

380,000 **736**

REPORTED 0.2% VALIDATION RATE

Generation now outruns validation. More candidates do not help if errors survive generation, prediction, synthesis, and validation.



Validation-rate indicator; not a universal synthesizability probability.

Lupine Science, “The 0.2% Synthesis Problem,” lines 7–16; Merchant et al., Nature 624 (2023), DOI 10.1038/s41586-023-06735-9.

Foundation MLIPs were measured across 21 materials, up to 9 properties, and 228 published reference values.

21

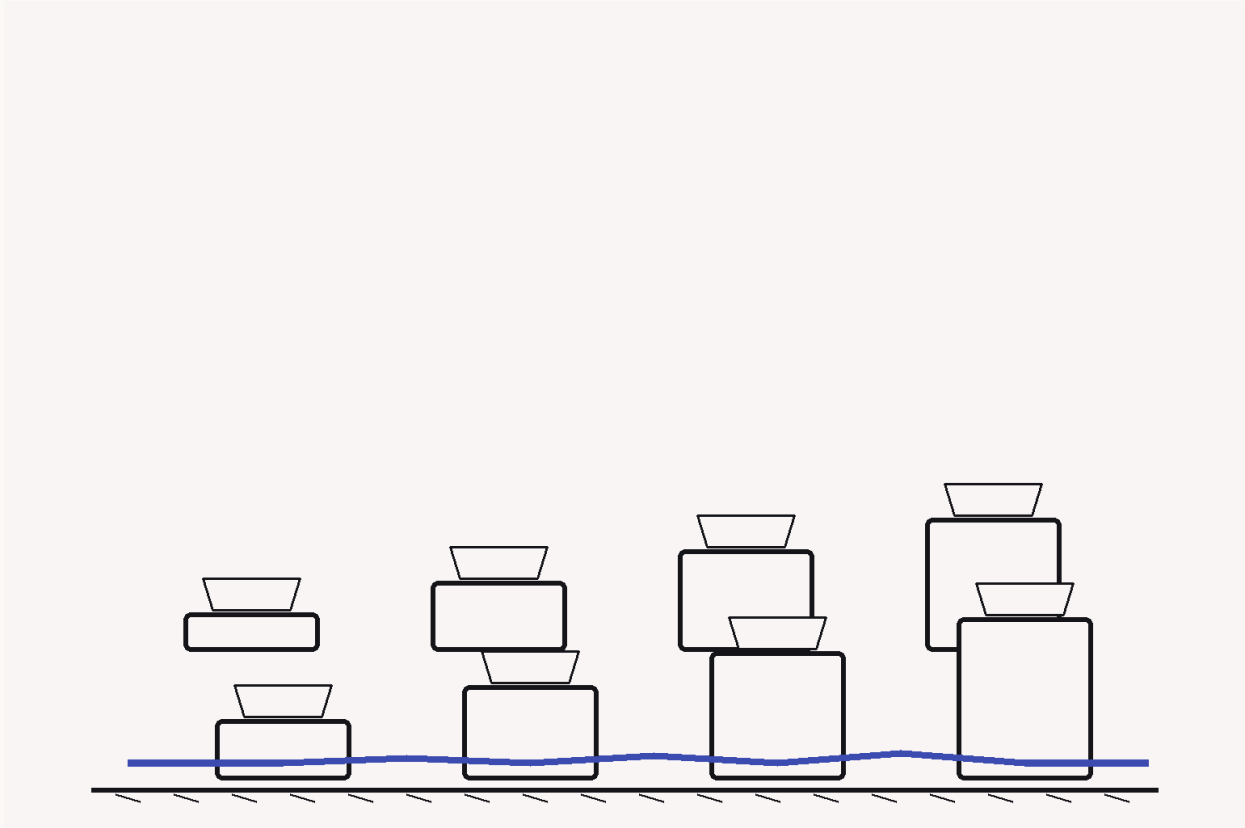
MATERIALS

up to 9

PROPERTIES

228

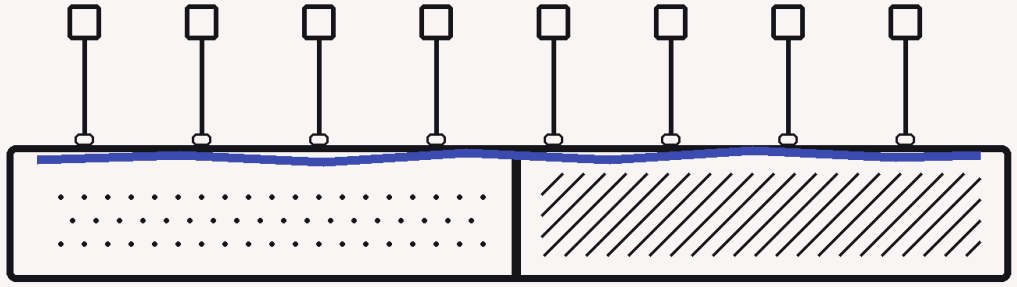
REFERENCE VALUES



In the measured benchmark, rankings often survive even when hard-property magnitudes miss by tens of percent. Bounded to the measured benchmark domain.

Lupine Science, “The Order Is Right, the Size Is Wrong,” lines 5–18 and 41–63.

Correction is a research question here, not a product. One narrow correction has held blind.



EXISTING MLIP

↓ MEASURED ROUTE

MEASURED ERROR FIELD

↓ ADDITIVE TERM

CORRECTED ENERGY + ANALYTIC FORCES

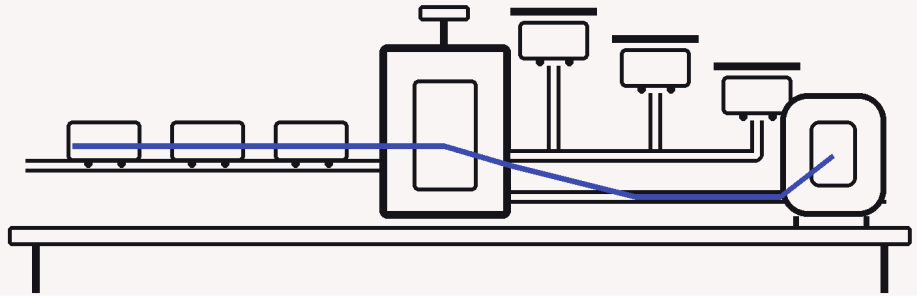
Beside CHGNet, blind (110) surface energy: Ni 9.7% → 1.5%; Cu 28.0% → 13.7%. One model, two metals, one facet.

Lupine Science, “A smooth, environment-resolved error field…,” line 27.

Every test ends in one of three verdicts against a gate written down in advance: pass, fail or abstain.

PASS	FAIL	ABSTAIN

In four preregistered campaigns in July 2026 (Round 4), no gate passed: three failed and one abstained. Our first correction made alloy predictions worse, and that record stays public too.



Amendments made after results are seen are labelled; the original verdict stays beside them.

Lupine Science, “Four Gates, Three Honest Failures…,” lines 14–60; “The Correction That Hurt…,” lines 19–35.

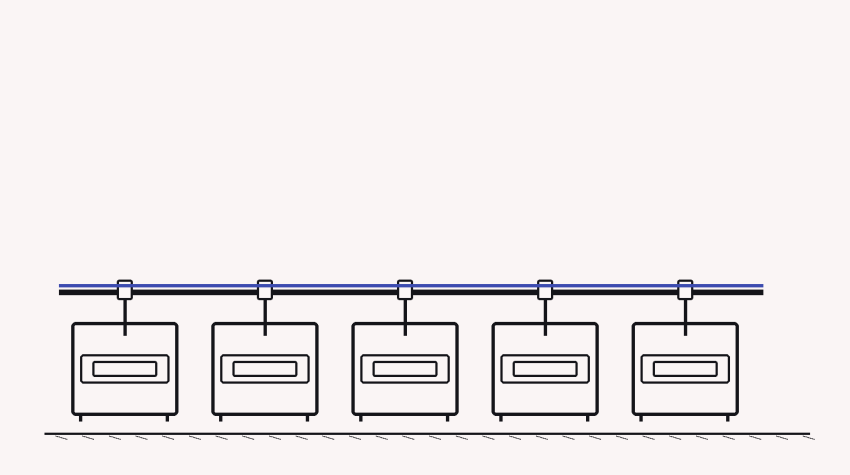
Executed: on a 30-path Li-ion barrier panel, four foundation MLIPs missed our 40 meV gate.

135.0–242.5 meV

MAE ON COMPLETED PATHS

40 meV

PREREGISTERED GATE · FAILED



The evidence program spans **42 published Li-ion electrolyte chemistries**: a **30-path frozen test panel** plus **12 disjoint training chemistries**, measured with MACE-MP-0 small and medium, MACE-MPA-0 medium and CHGNet.

Round 4, July 2026. Executed measurement; not peer-reviewed.

Lupine Science, “Four Gates, Three Honest Failures…,” lines 20–32; “The Z1 Barrier Panel,” lines 1–17 and 23–66.

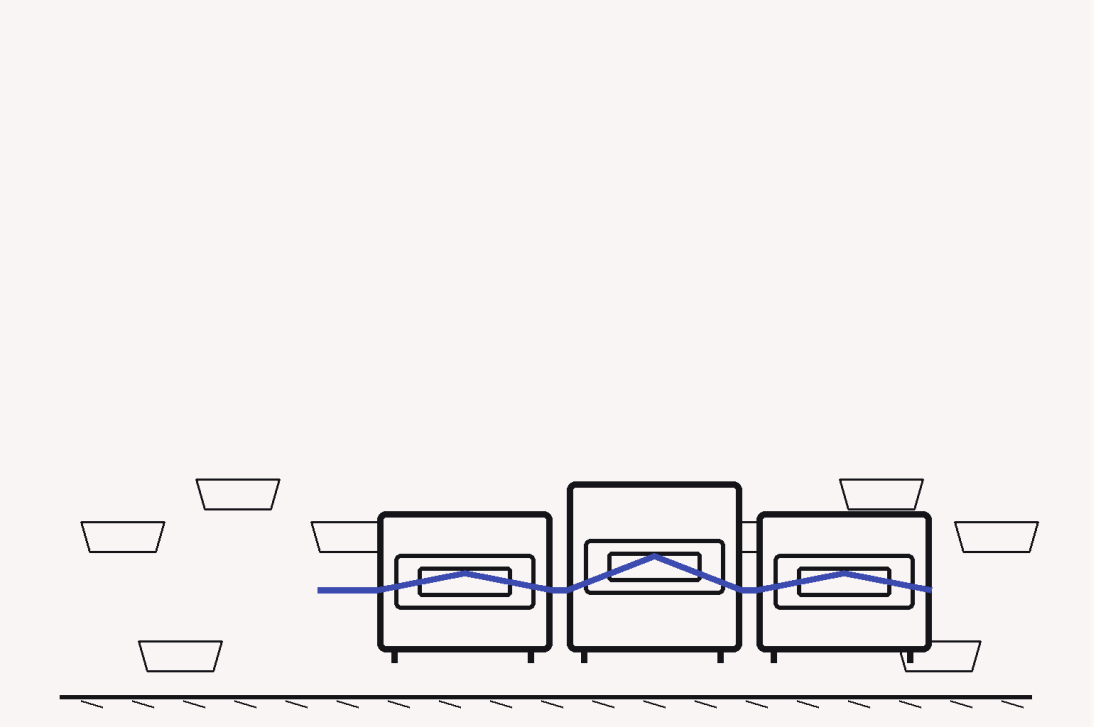
Projected, no new DFT: sharing anchors across four guided models would need 72.4% fewer DFT evaluations on 29 paths.

PROJECTED ACCOUNTING · NO NEW DFT

72.4% fewer

The baseline is four separate runs. Paths have 5–7 images, so the union covers most images and deduplication alone caps the saving near 73%.

PROJECTED, NOT MEASURED · NO EXTRAPOLATION



Lupine Science, “Shared DFT Anchors…,” lines 13–46; z1-union-anchor-economics.json (per-path image counts).

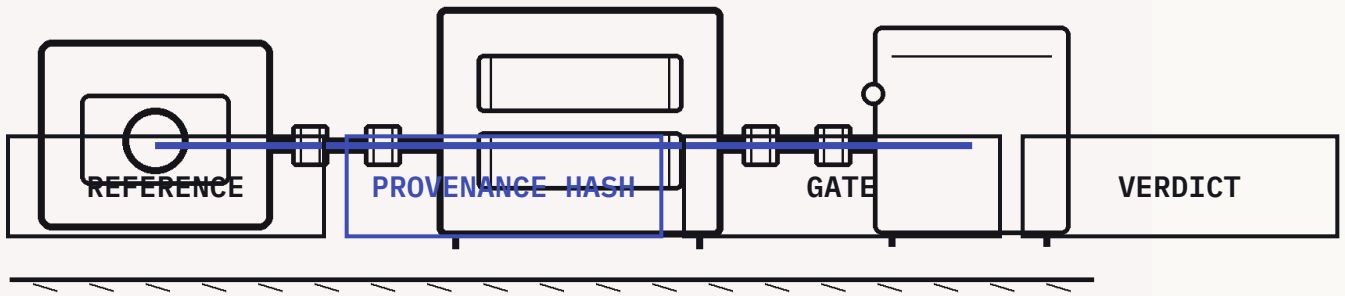
Every verdict keeps its record: the reference, the data hashes, the gate, any amendment, and the result.

Failed corrections stay in the record. Lean 4 checks the arithmetic and stated inequalities behind published numbers; it does not check the physics, the reference data or the pipeline.

Corrections log

Retired public claims are listed with dates and reasons.

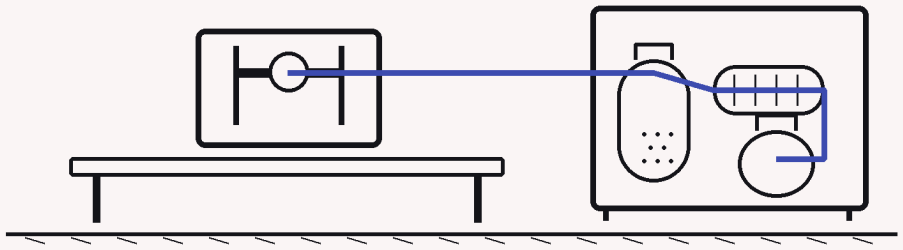
lupine.science/corrections/



Independent measurement between model builders and labs: builders get an outside check; labs see where a model has already failed.

MODEL GENERATION	INDEPENDENT MEASUREMENT	LABORATORY VALIDATION

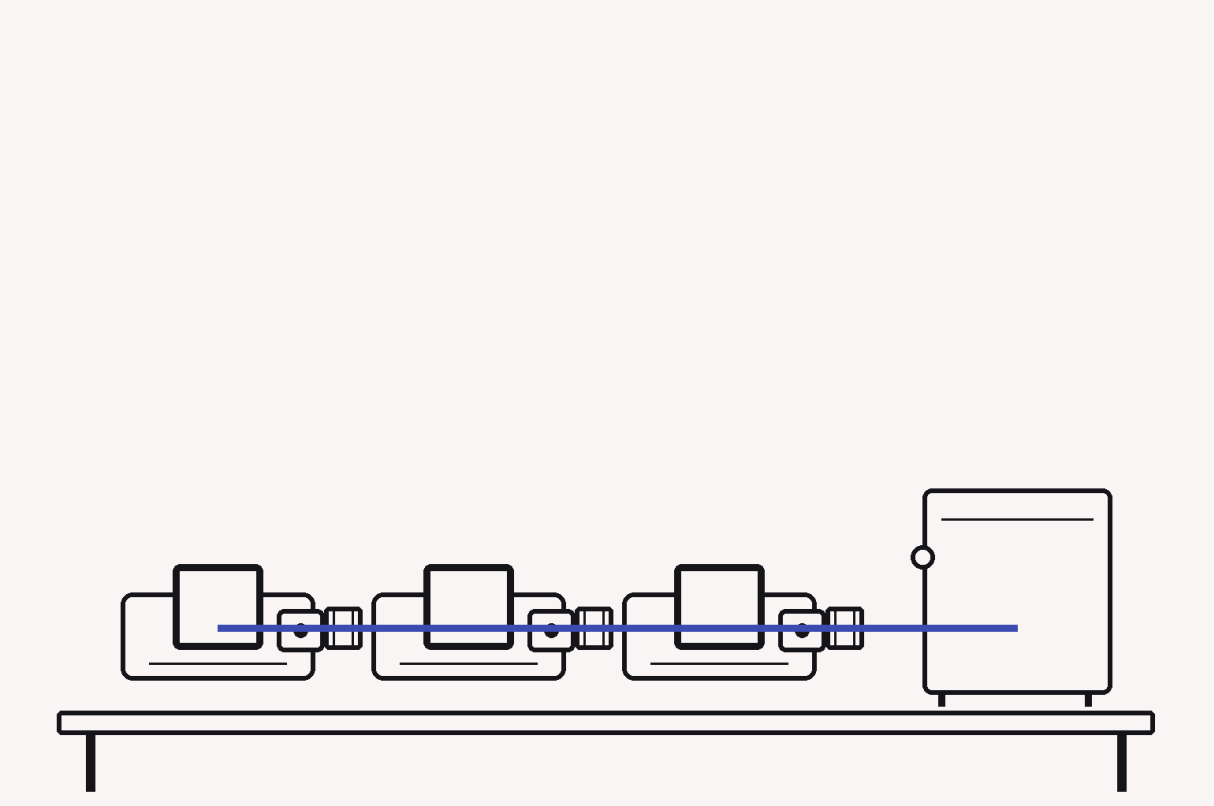
Initial market: teams already running MLIP simulation, across batteries, catalysts, frameworks, climate materials and alloys. We complement generators, databases, simulators and autonomous labs rather than replacing them.



Lupine Science, “The Trust Layer” (Direction: where the work is aimed), lines 64–78; company facts.

Inspectable now, each labelled for what it is: executed measurements, published failures, and one projected savings analysis.

frozen 30-path barrier panel	EXECUTED
Round-4 verdicts	EXECUTED
fcc field and elastic benchmarks	EXECUTED
shared-anchor savings	PROJECTED



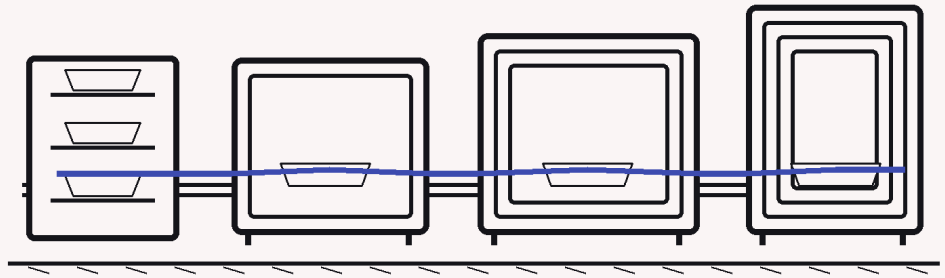
Research records — not commercial revenue, customers or peer-reviewed validation.

Lupine Science, “The Z1 Barrier Panel,” lines 40–66; “Four Gates, Three Honest Failures…”; “The Small Cell Held”; “Shared DFT Anchors…,” lines 13–46.

Advance by order of effort: statics → elastic response → surfaces → defects → transition states → interfaces → abstention-only domains.

STATICS	ELASTIC RESPONSE
SURFACES	DEFECTS
TRANSITION STATES	INTERFACES
ABSTENTION-ONLY DOMAINS	

Measure first where references are solid; extend to harder families only behind gates set in advance; use sparse DFT anchors where no correction holds; abstain where no supported measurement path exists.



Lupine Science, “An Order of Effort,” lines 19–55.

One 30-path panel. One chemistry family. Not peer-reviewed.

– the preregistered 40 meV gate against published references failed; the passing check is same-engine only, and against the references the pilot was 693.0–705.7 meV off

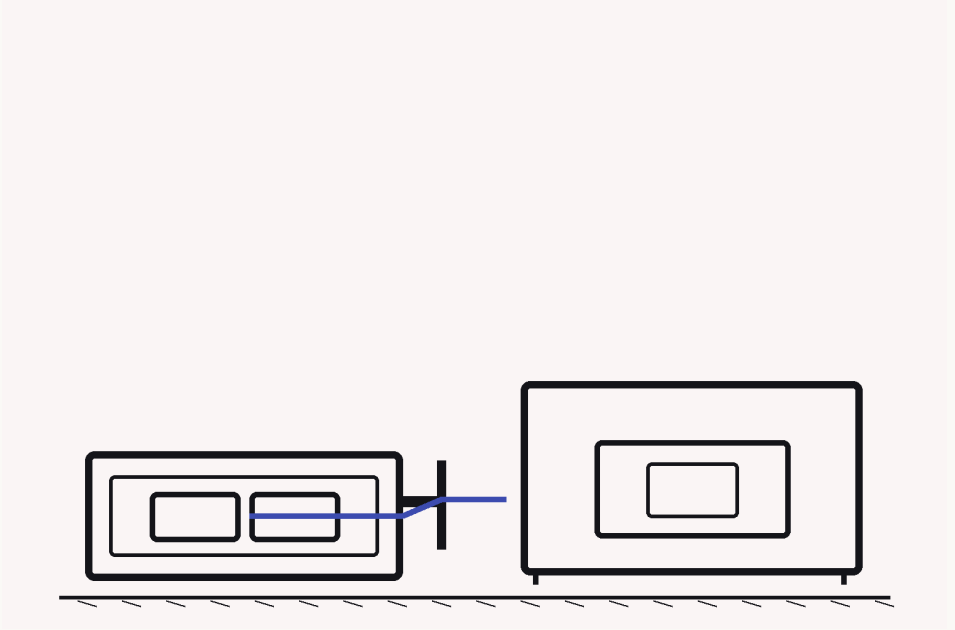
– one DFT engine and functional in the executed Z1 record

– short paths make sparsity easier

– seven large paths remain deferred

– two models recorded guidance misses

– no GPAW↔VASP equivalence claim



Lupine Science, “The Savings Stack,” lines 38–42; “Shared DFT Anchors…,” lines 80–90; Z1 de-brief (archived), lines 31–36.

Make measured model error part of the simulation loop materials labs already run.

[OWNER DECISION] Round size
[OWNER DECISION] Instrument / terms
[OWNER DECISION] Runway and allocation

Models guide. Evidence measures. Lean checks the arithmetic.

USE-OF-FUNDS FRAME

Measurement on the lanes we have tested.

Gated expansion across harder structure and property families.

Evaluation that runs on a lab's own hardware, from simulation to synthesis.

